

Use the following values, unless specified.

For 0.25 μ m Technology node: $V_{DD}=2.5$ V, $V_{T0N}=0.4$ V, $V_{T0P}=-0.4$ V, $L_{min}=0.25$ μ m, $C_{OX}=6$ fF/ μ m², $\lambda_n=0.05$ V⁻¹, $\lambda_p=0.05$ V⁻¹, $\mu_n C_{OX}=120$ μ A/V², $\mu_p C_{OX}=40$ μ A/V².

For Silicon: $\epsilon_{Si}=11.8 \epsilon_0$, $\epsilon_0=8.85 \times 10^{-14}$ F/cm.

For digital gates: $R_{nmos}=13$ k Ω / (W/L), $R_{pmos}=31$ k Ω / (W/L).

For short channel MOSFET (additional parameters): $v_{sat}=10^5$ m/sec, $V_{DSATn}=0.63$ V, and $V_{DSATp}=-1$ V

Q.1. Answer the following.

- a. In the OP-AMP topology given in **Fig. 1 (a)**, find the **expressions and values** of small signal differential gain $[V_{OUT}/V_{IN}]$, Input Common Mode Range (ICMR) and output voltage swing. Assume that all the transistors are in saturation, and $I_{SS}=20$ μ A. Take, for each transistor, $W/L=10$.
- b. Find the **H-parameters** of the two-port network shown in **Fig. 1 (b)**. [20]

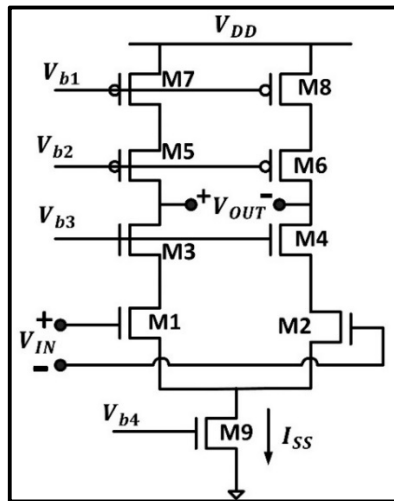


Fig. 1 (a)

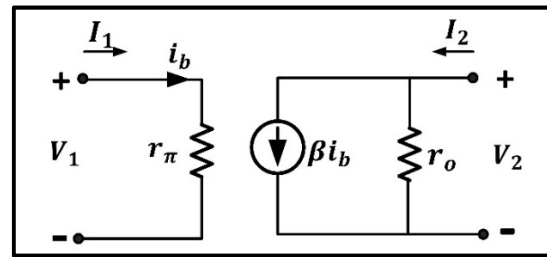


Fig. 1 (b)

Q.2. Answer the following.

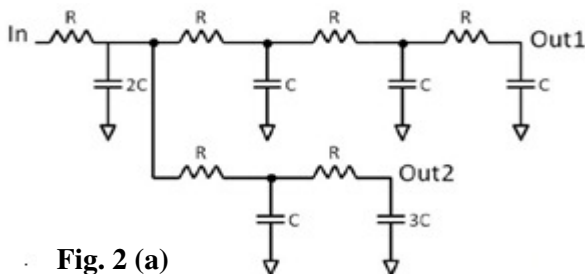


Fig. 2 (a)

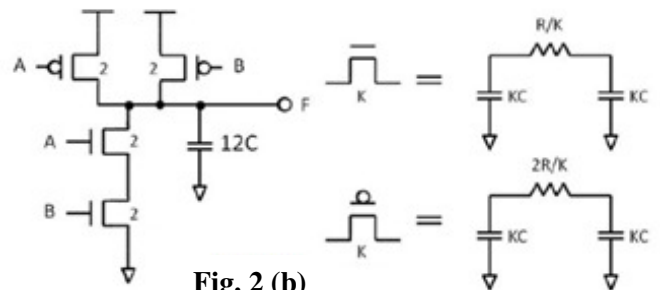


Fig. 2 (b)

Fig. 2 (a): An RC network; **Fig. 2 (b):** 2-input NAND gate and corresponding NMOS and PMOS transistor models.

P.T.O.>>

Note: In delay calculation, use ‘ln(2)’ expression with $R = 100\ \Omega$ and $C = 10\ \text{fF}$.

- Given the RC network in **Fig. 2 (a)**, calculate the Elmore delay from **In to Out1** and from **In to Out2**. Which one is critical path?
- Calculate the **T_{PLH} and T_{PHL}** of a two input NAND gate shown in **Fig. 2 (b)** using logical effort assuming $\mu_n:\mu_p=2:1$. Take $g_{inv}=1$, $p_{inv}=1$
- Calculate the delay in worst case for **T_{PLH} and T_{PHL}** of the same two input NAND gate shown in **Fig. 2 (b)** using Elmore delay.
- Is the delay between part (b) and (c) different and why? Explain in one statement.

[20]

Q.3. The digital system shown in **Fig. 3** below has two sequential elements (Edge triggered) *Reg-A* and *Reg-B* and two combinational logic blocks *forward path* and *backward path*. The timing parameters are mentioned in the tables adjacent to the figure. CLOCK signal can be launched at either of the ends *Node-A* or *Node-B* and *Delay* indicates the interconnect delay.

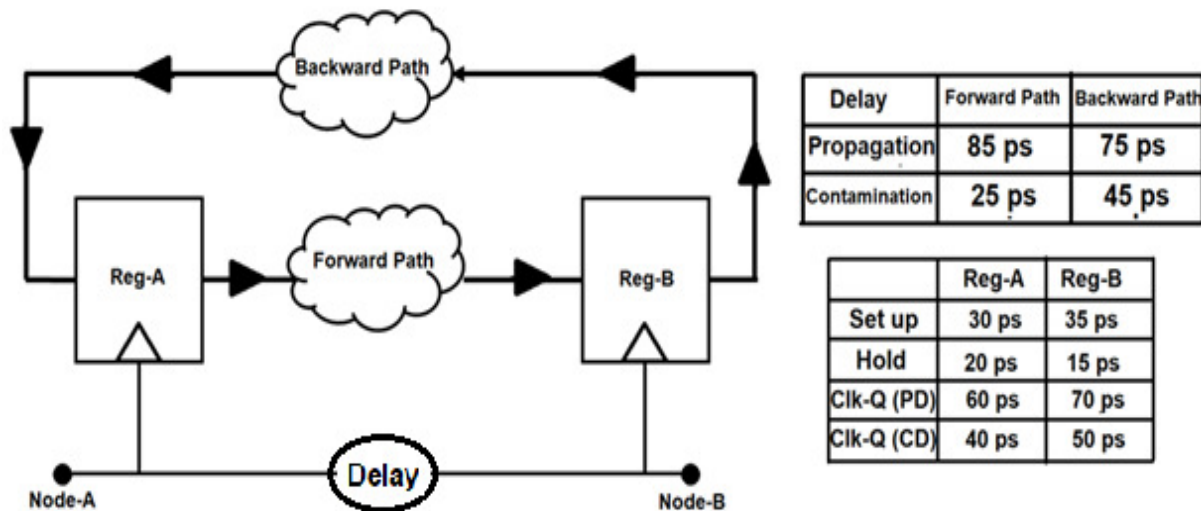


Fig. 3

- If CLOCK is launched at Node-A and Delay=25 ps. What is the maximum CLOCK frequency at which the system can be operated?
- What is the maximum tolerable value of *Delay*? For this you may have to consider two cases, CLOCK being launched at Node-A and Node-B separately.
- What is advisable? Launching CLOCK from Node-A or Node-B? Justify. Clearly state the reasons.

[15]

----- END -----

Date: 13/12/2019 FN
[Total Time: 90 min]

(Part B—Open book)

Part B
Max. Marks: 55

General Instructions to the candidate

- Please write your answers legibly and neatly in the answer booklet provided only.
- Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of three questions and all are compulsory. Each carries marks as indicated.

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μm Technology node— $V_{DD}=2.5\text{ V}$, $V_{T0N}=0.4\text{ V}$, $V_{T0P}=-0.4\text{ V}$, $L_{\min}=0.25\text{ }\mu\text{m}$, $C_{OX}=6\text{ fF}/\mu\text{m}^2$, $\lambda n=0.05\text{ V}^{-1}$, $\mu_n C_{OX}=120\text{ }\mu\text{A/V}^2$, $\mu_p C_{OX}=40\text{ }\mu\text{A/V}^2$. For Silicon: $\epsilon_{Si}=11.8\epsilon_0$, $\epsilon_0=8.85\times 10^{-14}\text{ F/cm}$.
For digital gates— $R_{\text{onmos}}=13\text{ kohm}/(\text{W/L})$, $R_{\text{pnmos}}=31\text{ kohm}/(\text{W/L})$.

For short channel MOSFET (additional parameters)— $v_{\text{sat}}=10^5\text{ m/sec}$, $V_{\text{DSATN}}=0.63\text{ V}$, $V_{\text{DSATP}}=-1\text{ V}$.

Unless specifically mentioned—

- Use long channel approximation and lambda based design rules unless specially mentioned.
- Use square law current equation. All dimensions are in ' μm ' units.
- Neglect body effect, channel length modulation if not mentioned specifically. Assume step input/s

The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

- Q.1. Consider the circuit shown in Fig. Q.1 $V_{in1}-V_{in2}=V_{in}=10\text{ mV}$. Under matched conditions, dc current $I_{M1}=I_{M2}=I_{SS}/2=50\text{ }\mu\text{A}$. Given $C_L=5\text{ pF}$ and $V_{ov}=0.2\text{ V}$ for all transistors. Assume matched transistors in differential arms and impedance r_{o7} is very high. All transistors operate in active region.

Given dc current $I_1=I_2=I_{SS}/2$, $I_3=I_4=I_{SS}/8$, $I_5=I_6=3I_{SS}/8$

- Find values of OCMR (output common mode range), output signal swing and pole frequency magnitude due to output node.
- Determine magnitude of G_m , R_{out} and Open loop voltage gain $A_v=[V_{out}/V_{in}]$ at low frequency.
- Now, if the bias voltages V_{b21} and V_{b22} are varied to increase I_5 by $I_{SS}/8$ and decrease I_6 by $I_{SS}/8$, fill the entries in the Table 1. For this part, assume $r_{o7}\rightarrow\infty$.

Now using two configurations of this differential amplifier, design the schematic of a variable gain amplifier with V_{b21} and V_{b22} as control voltages to change currents I_5 and I_6 , simultaneously, as given below—

I_5 from $4(I_{SS}/8)\rightarrow(3I_{SS}/8)\rightarrow(2I_{SS}/8)$, and I_6 from $(2I_{SS}/8)\rightarrow(3I_{SS}/8)\rightarrow(4I_{SS}/8)$.

[20]

Table -1

Parameter	Value [for Part (a)]	Value [for part (c)]
I_5		
I_6		
I_3		
I_4		
I_1		
I_2		
I_7		
g_{m1}		
g_{m2}		
R_{out}		

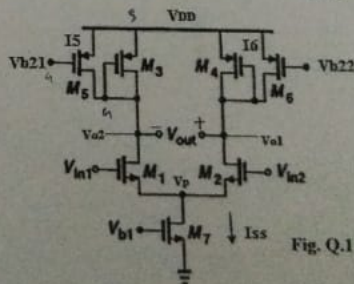


Fig. Q.1

P.T.O.

Shen...

Q.2. A two-stage single-ended operational amplifier with basic current-mirror load has a dominant pole p_1 due to miller compensation, pole p_2 at output node and a mirror pole p_3 . It also has a right half plane (RHP) zero z . Assume DC voltage gain A_0 to be high.

- Write down the transfer function of opamp.
- If RHP zero is at 15GBW, pole p_3 at 4GBW, then find out location of pole p_2 corresponding to 55° phase margin. Take Gain bandwidth (GBW) to be 10 MHz.
- Now consider the case when miller compensation is modified by the use of nulling resistance. Therefore, the location of RHP zero is changed such that it overlaps with pole p_2 at LHP. Assuming, addition of nulling resistance does not generates any other additional poles and location of pole p_1 and p_3 remains same as of case ii, find out the phase margin.

[15]

Q.3. For Bandgap reference circuitry shown in Fig. Q.3, assume all the resistances are independent of temperature and opamps to be ideal. Emitter area of BJTs Q_1 and Q_2 are in the ratio: $A_{E1} : A_{E2} = 1 : 10$. Transistors M_1, M_2, M_3 are perfectly matched and have equal W/L ratio.

- Find out simplified expression for V_{REF} as a function of V_{EB3} .
- Find out the ratio $R_2 : R_1$ for zero temperature coefficient (ZTC) of reference voltage (V_{REF}).
- If $R_1 = R = 1 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$, find out temperature coefficient of V_{REF} in PPM/ $^\circ\text{C}$.

Given: $V_{EB3} = 0.7 \text{ V}$, $\frac{\partial V_{EB}}{\partial T} = -1.8 \text{ mV}/^\circ\text{C}$ and $\frac{\partial V_T}{\partial T} = +0.087 \text{ mV}/^\circ\text{C}$, $V_T = 26 \text{ mV}$, where V_T is thermal voltage.

[20]

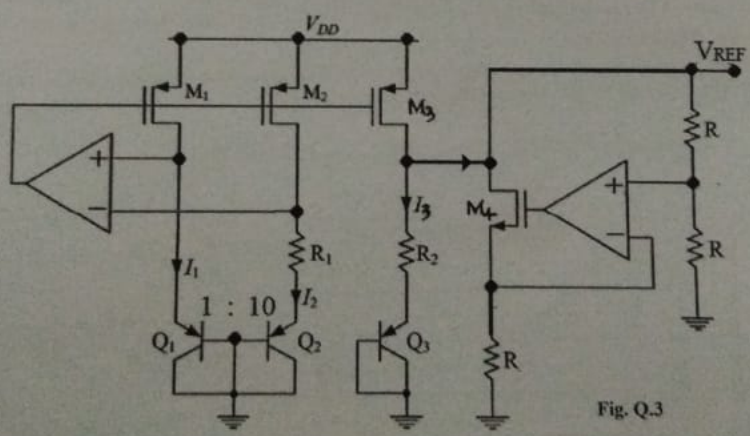


Fig. Q.3